

Monster Lab

Build Git, one problem at a time



Open: _____

Type your name.

Pick your lab.

Lab 1

Lab 2

Lab 3

QR

STEP 1**Save every version**

Why do arrows point back, never forward?

git commit

Each save makes a new card that never changes.

BEHIND THE DOOR

`commit = tree + parent + author + time + message -> SHA-1 ID`

STEP 2**Try two ideas at once**

Where is the original monster now? Did anything get copied?

git switch -c cat-robot

A sticky note is a label, not a copy.

BEHIND THE DOOR

`.git/refs/heads/cat-robot` holds one commit ID · HEAD points to it

STEP 3**Make one monster from both**

Why did FACE and LEGS combine alone, but BODY needed you?

git merge superhero

Compare both sides with the card they share.

BEHIND THE DOOR

3-way merge vs merge-base · both changed, differently -> CONFLICT

STEP 4

Meet the Wall

Two labs never made that card. Why does their copy have the same ID?

git clone

Every lab has a full copy with the same IDs.

BEHIND THE DOOR

ID = SHA-1 of the commit's bytes · same bytes -> same ID everywhere

STEP 5**Put your monster on the Wall**

Why did the Wall refuse your card instead of adding it?

git push • git pull

The Wall only moves forward, so combine first.

BEHIND THE DOOR

push: fast-forward only, else ! [rejected] • pull = fetch + merge

STEP 6**Oops: undo a shared mistake**

Why is adding a fix card safe, but moving back is not?

git revert

Don't rip out a shared card; add a fix card.

BEHIND THE DOOR

revert adds an inverse commit · reset moves the branch · reflog logs it

STEP 7**The boss wants it clean**

Who added the tentacles? Where does that answer still exist?

git push --force

Force the Wall, and it forgets who did what.

BEHIND THE DOOR

squash: new ID · --force: no fast-forward check · gc: old commits gone

Switching to Git: Good, Bad, Ugly

GOOD

Cheap branches.

Local commits and reverts.

Developers prefer flat history.

BAD

Fast-forward: no merge commit, branch forgotten.

Rebase: the change itself is rewritten.

Squash: commits gone, even who made them.

UGLY

Integration path: a change's route to main.

Microsoft rebuilt its tracing for Git.

Some loss can't be recovered.

Flat history is data loss.

You saw it in Step 7: squash, then force push. The Wall forgot who added 🐙.

Cards never change.
Sticky notes move.
The Wall copies cards.